Prompt-Level Grounding Connects LLM Ideas to Literature but Does Not Improve Judged Quality

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Abstract

Large language models can write plausible scientific ideas, but scientific ideation requires more than plausible novelty claims: ideas must be different from prior work and realistic enough to test. We examine whether novelty improves when it is grounded through concrete prompt operations rather than requested abstractly. Using 12 matched AI IDEA BENCH 2025 contexts, we generate 84 ideas with GPT-4.1-MINI under seven conditions: an UNGROUNDED baseline plus deduction, induction, proposal writing, outcome imagination, a small lexical audit, and explicit literature comparison. We evaluate each idea with an independent GEMINI-2.5-FLASH judge on novelty, feasibility, specificity, grounding, risk awareness, and overall quality, and we add automatic overlap and diversity metrics. The result is mixed-to-negative. LITERATURE COMPARISON has the highest mean overall score, 4.00 versus 3.92 for UNGROUNDED, but no judged quality metric shows a significant condition effect; for overall score, the Friedman test gives $p = 0.157$. The UNGROUNDED baseline has the best novelty–feasibility product, 14.67. Grounded prompts do reliably change surface relation to prior work: prior-work overlap increases for every grounded condition, with a strong omnibus effect ($p = 1.06 \times 10^{-9}$). These results suggest that prompt-level grounding alone makes ideas sound more literature-connected without reliably improving idea quality.

1 Introduction

Scientific idea generation is useful only when ideas are both new and executable. Large language models (LLMs) can produce fluent research proposals, name plausible datasets, and write convincing novelty claims. This fluency creates a practical problem: a generated idea can look grounded in a field while still being a close recombination of prior work or too underspecified to run.

Recent systems therefore move beyond direct brainstorming. They retrieve related papers, compare candidate ideas against prior work, organize literature trajectories, rank proposals, or even execute experiments [Wang et al., 2024a, Baek et al., 2024, Wang et al., 2024b, Li et al., 2024, Lu et al., 2024]. This shift is important, but it leaves a simple controlled question open. If we hold the topics, generator, judge, and evaluation rubric fixed, do different grounding operations improve generated scientific ideas over a direct prompt for novelty?

Our main question. We ask whether LLM-generated scientific ideas improve when novelty is grounded through explicit reasoning operations or evidence. We focus on six prompt-level grounding operations: deduction from prior-work titles, induction from apparent title trends, compact proposal writing, imagined positive and negative outcomes, a small empirical audit over prior-work title terms, and explicit literature comparison. We compare all six against an UNGROUNDED baseline that receives only a topic and keywords.

We run a matched pilot on 12 AI IDEA BENCH 2025 contexts [Qiu et al., 2025]. For each context and condition, GPT-4.1-MINI generates one idea through OpenRouter [OpenRouter, 2026]. A different model, GEMINI-2.5-FLASH, judges every idea on novelty, feasibility, specificity, grounding,

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risk awareness, and overall quality. We complement these judged scores with a lexical prior-work overlap metric and within-condition TF-IDF diversity.

The main finding is cautionary. LITERATURE COMPARISON obtains the highest mean overall score, 4.00 compared with 3.92 for UNGROUNDED, but this difference is tiny and not statistically reliable. Across all seven conditions, no judged quality metric has a significant Friedman omnibus effect: overall quality has $p = 0.157$, quality mean has $p = 0.208$, and novelty–feasibility product has $p = 0.572$. The strongest reliable effect is lexical: all grounded conditions increase prior-work overlap relative to UNGROUNDED after Holm correction, and LITERATURE COMPARISON has the largest increase ($+0.063$, paired Cohen's $d = 2.84$, Holm $p = 0.0029$).

In summary, our main contributions are:

- We conduct a matched comparison of six prompt-level grounding operations against an UNGROUNDED novelty baseline on identical AI research contexts.
- We evaluate generated ideas with an independent LLM judge and automatic metrics for prior-work overlap and diversity.
- We show that simple grounding increases visible connection to prior-work titles but does not significantly improve judged novelty, feasibility, specificity, overall quality, or novelty–feasibility product in this pilot.
- We identify practical design implications for future ideation systems: stronger retrieval, anti-copying objectives, candidate diversification, human calibration, and execution feedback.

Section 2 reviews LLM scientific ideation and novelty evaluation. Section 3 describes the matched experimental design. Section 4 reports the quantitative results. Section 5 interprets the findings, and section 6 summarizes next steps.

2 Related Work

Literature-grounded idea generation. Early LLM ideation studies showed that models can generate plausible research ideas, but the best systems increasingly add literature grounding. SCIMON retrieves inspirations from prior papers and iteratively revises ideas to reduce overlap with literature [Wang et al., 2024a]. RESEARCHAGENT uses scientific literature and reviewing agents to iteratively generate research problems, methods, and experiments [Baek et al., 2024]. SCIPIP improves retrieval with semantic, citation, and full-text information, then combines retrieved content with LLM internal knowledge [Wang et al., 2024b]. CHAIN OF IDEAS organizes papers into chains that model how research directions develop over time [Li et al., 2024]. Unlike these systems, we do not propose a full ideation agent. We isolate prompt-level grounding operations under matched topics, model, and evaluation criteria.

Evaluation of scientific ideas. Evaluation is the central challenge for automated ideation. Si et al. [2024] ran a controlled human study with more than 100 NLP researchers and found that LLM ideas can be judged more novel than expert ideas while being weaker on feasibility and diversity. RINOBENCH turns expert review data into a benchmark for novelty judgment and shows that LLM rationales can resemble expert rationales while numeric novelty scores still diverge from human labels [Schopf and Färber, 2026]. The Idea Novelty Checker frames novelty assessment as a literature-grounded retrieval and reranking problem [Shahid et al., 2025]. We use an independent LLM judge because the present experiment evaluates newly generated ideas, but we report automatic overlap and paired tests to avoid relying only on one scalar score.

Idea-generation benchmarks. IDEABENCH and AI IDEA BENCH 2025 provide structured paper and reference contexts for research idea generation [Guo et al., 2024, Qiu et al., 2025]. These benchmarks are useful because they separate a research context from a target paper idea. We use AI IDEA BENCH 2025 contexts as input topics, but we do not expose hidden target-paper methods, motivations, or summaries to the generator. Our goal is not to recover the target paper; it is to estimate how prompt grounding changes the quality and literature proximity of generated ideas.

Execution as grounding. A stronger form of grounding is to run experiments. MLAGENTBENCH evaluates agents on machine-learning experimentation tasks [Huang et al., 2023]; DISCOVERYBENCH formalizes data-driven discovery tasks [Majumder et al., 2024]; and SCIENCEAGENTBENCH tests scientific agents on validated coding tasks from scientific workflows [Chen et al.,

Condition	Role	Prompt operation
UNGROUNDED	Baseline	Generate a novel AI research idea from the broad topic and keywords only.
DEDUCTION	Grounded	Infer a gap from topic, keywords, and prior-work titles, then propose an idea for the gap.
INDUCTION	Grounded	Infer a trend from prior-work titles and propose the next plausible research step.
PROPOSAL	Grounded	Write a compact proposal with method, data, baselines, metrics, and ablations.
OUTCOME IMAGINATION	Grounded	Imagine positive and negative outcomes before finalizing the idea.
SMALL EXPERIMENT	Grounded	Use a real lexical metadata audit over prior-work titles before proposing the idea.
LITERATURE COMPARISON	Grounded	Compare against each prior-work title and revise the idea to differentiate it.

Table 1: Matched prompt conditions. Every context receives one generated idea under each condition.

2024]. THE AI SCIENTIST combines idea generation, code execution, paper writing, and simulated review [Lu et al., 2024]. Our SMALL EXPERIMENT condition is much weaker: it uses a real but shallow lexical audit over titles. The negative result for this condition supports the broader lesson that grounding needs meaningful evidence, not just an empirical-looking step.

3 Methodology

Problem setup. Let c_i be an AI research context and k be a prompt condition. A generator model produces one idea y_{ik} for every pair (c_i, k). We then score each idea with a judge model and compute automatic metrics from the idea text and the context's prior-work titles. Because every context receives every condition, all comparisons are paired by context.

Data. We sample 12 contexts from the local AI IDEA BENCH 2025 test split with seed 42. Each context contains a broad revised topic, keywords, and prior-work titles. The generator sees the topic and keywords in every condition and sees prior-work titles only when the prompt condition requires them. We hide target paper names, target methods, target motivations, and target summaries from the generator.

Prompt conditions. Table 1 lists the seven matched conditions. The primary baseline is UNGROUNDED, which asks directly for a novel idea from the topic and keywords. The six grounded conditions add one operation while keeping the topic distribution, generator, temperature, and output requirements fixed.

Generation and judging. We use GPT-4.1-MINI as the generation model through OpenRouter chat completions [OpenRouter, 2026]. Generation temperature is 0.7. We use GEMINI-2.5-FLASH as the independent judge at temperature 0.0. The raw generations and evaluations are saved in `results/model_outputs/generations.jsonl` and `results/model_outputs/evaluations.jsonl`. The experiment generates 84 complete idea/evaluation pairs across 12 contexts and 7 conditions.

Judged metrics. The judge scores each idea from 1 to 5 on novelty, feasibility, specificity, grounding, risk awareness, and overall quality. Novelty measures conceptual distance from prior work; feasibility measures whether the study is executable with plausible data, baselines, and resources; specificity measures concrete method and evaluation detail; grounding measures whether the idea uses evidence or constraints without merely copying prior work; and overall measures scientific promise. We define quality mean as the mean of novelty, feasibility, specificity, grounding, and overall. We also report the novelty–feasibility product, novelty $\times$ feasibility, on a 1–25 scale.

Automatic metrics. We compute prior-work overlap as the fraction of generated content terms that also appear in the context's prior-work titles. This metric is intentionally simple: it measures visible lexical connection to the provided titles, not semantic copying. We also compute a within-condition diversity score as mean pairwise TF-IDF cosine distance among ideas in the same condition.

Condition	n	Novelty	Feas.	Spec.	Ground.	Overall	Nov.×Feas.	Prior Overlap
UNGROUNDED	12	**3.75**	3.92	**3.92**	3.92	3.92	**14.67**	**0.043**
DEDUCTION	12	3.50	3.83	3.67	4.00	3.75	13.33	0.077
INDUCTION	12	3.42	**4.00**	3.75	4.00	3.75	13.67	0.092
PROPOSAL	12	3.58	3.92	**3.92**	4.00	3.92	14.00	0.073
OUTCOME IMAGINATION	12	3.50	**4.00**	3.75	4.00	3.83	14.00	0.059
SMALL EXPERIMENT	12	3.33	**4.00**	3.58	4.00	3.58	13.33	0.065
LITERATURE COMPARISON	12	**3.75**	3.83	3.75	**4.08**	**4.00**	14.33	0.106

Table 2: Mean scores by condition. Judged scores are on a 1–5 scale. Higher is better except prior-work overlap, where lower means less lexical reuse from prior-work titles. Best values are in **bold**.

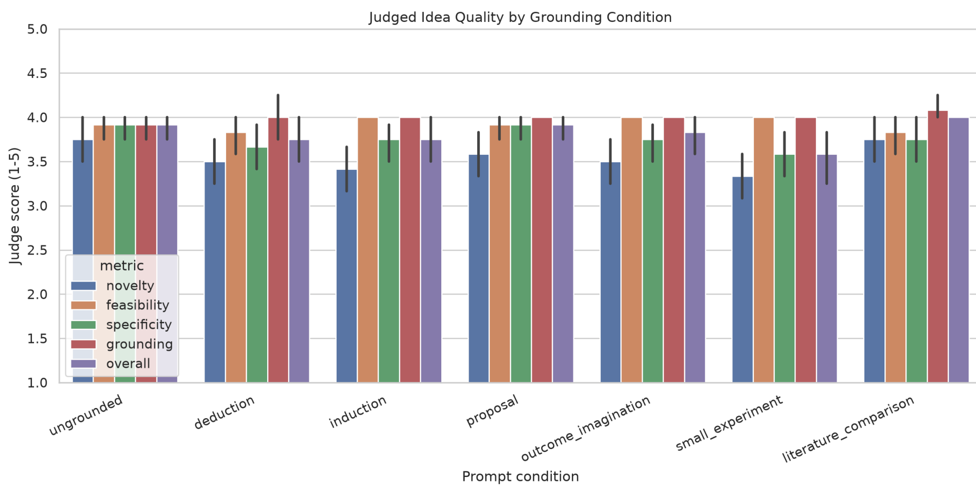

Figure 1: Judged idea quality by prompt condition. The grounded prompt variants do not separate cleanly from the UNGROUNDED baseline on novelty, feasibility, specificity, grounding, or overall quality.

Statistical analysis. We use Friedman tests for omnibus condition effects because each context appears under every condition. For pairwise comparisons, we compare each grounded condition against UNGROUNDED with Wilcoxon signed-rank tests and Holm correction. We report paired Cohen's d as an effect size and bootstrap 95% confidence intervals for condition means. We interpret statistical tests cautiously because this is a compact pilot with $n = 12$ contexts.

4 Results

Main judged scores. Table 2 summarizes all judged and automatic metrics by condition. The strongest grounded condition by overall quality is LITERATURE COMPARISON, with mean overall 4.00 compared with 3.92 for UNGROUNDED. This is a small absolute gain of 0.08 points on a 1–5 scale. The UNGROUNDED baseline remains strongest on the novelty–feasibility product, with a mean of 14.67 compared with 14.33 for LITERATURE COMPARISON and 14.00 or lower for the other grounded conditions.

Figure 1 shows the judged score distributions by condition. The main visual pattern is compression: most scores fall between 3 and 4, with limited separation among prompt variants. This ceiling makes large gains unlikely and makes paired tests important.

Omnibus tests. Table 3 reports Friedman tests across all seven conditions. No judged quality metric shows a significant condition effect. Overall quality has $p = 0.157$, quality mean has $p = 0.208$, and novelty–feasibility product has $p = 0.572$. Grounding score itself also does not significantly differ across conditions ($p = 0.701$), despite grounded prompts being more explicit about prior work.

Prior-work overlap changes sharply. The one strong condition effect is prior-work overlap. The Friedman statistic is 53.21 with $p = 1.06 \times 10^{-9}$. Every grounded condition has higher overlap

Metric	Friedman statistic	p-value
Novelty	8.60	0.197
Feasibility	7.29	0.295
Specificity	6.59	0.361
Grounding	3.82	0.701
Overall	9.32	0.157
Quality mean	8.44	0.208
Novelty–feasibility product	4.78	0.572
Prior-work overlap	**53.21**	1.06×10^{-9}

Table 3: Omnibus paired tests across all seven conditions and 12 matched contexts. Only prior-work overlap shows a significant condition effect.

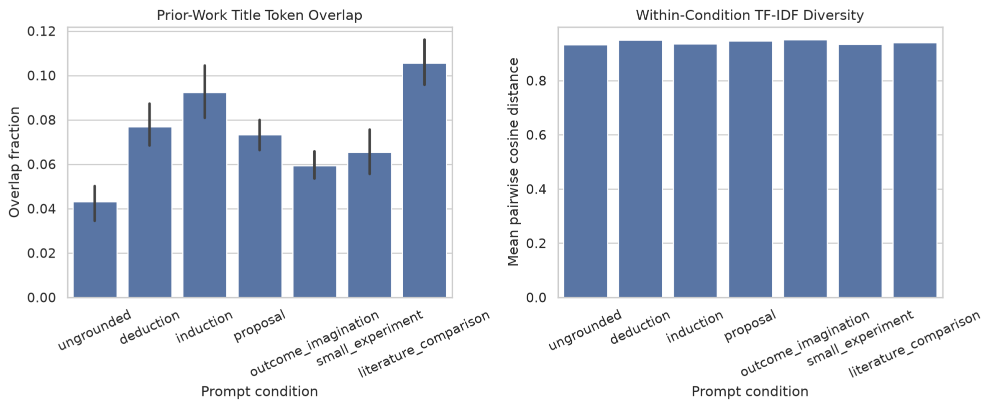

Figure 2: Prior-work overlap and within-condition diversity. Grounded prompts consistently increase lexical overlap with prior-work titles, with the largest increase under LITERATURE COMPARISON. Diversity remains high across all conditions and does not explain the main judged-score pattern.

than UNGROUNDED after Holm correction. LITERATURE COMPARISON has the largest increase over baseline: $+0.063$ in mean overlap, paired Cohen's $d = 2.84$, and Holm-adjusted $p = 0.0029$. Figure 2 shows this pattern together with condition diversity.

Novelty–feasibility tradeoff. Figure 3 focuses on the novelty–feasibility product. If prompt grounding improved the practical novelty of ideas, we would expect grounded conditions to exceed UNGROUNDED on this product. Instead, UNGROUNDED has the highest mean product, 14.67, and all grounded conditions are lower. Pairwise differences versus UNGROUNDED are not significant after Holm correction, but the direction is important: prompt-level grounding did not improve the intended novelty–feasibility tradeoff.

Ablation interpretation. The condition comparison acts as an ablation over grounding operations. PROPOSAL matches UNGROUNDED on specificity and overall score but does not improve novelty. OUTCOME IMAGINATION keeps feasibility high but does not improve overall quality. SMALL EXPERIMENT performs worst on novelty, specificity, overall score, quality mean, and novelty–feasibility product. The small audit used only title-term frequencies and missing evaluation cues, so this result suggests that empirical grounding must provide meaningful evidence before it can improve idea quality.

Qualitative errors. Error analysis shows repeated underspecification. Weak ideas often rely on vague mechanisms such as "adaptive modules," "contrastive heads," or unspecified feedback loops. Feasibility scores remain high because the generator can name datasets, baselines, and metrics, but

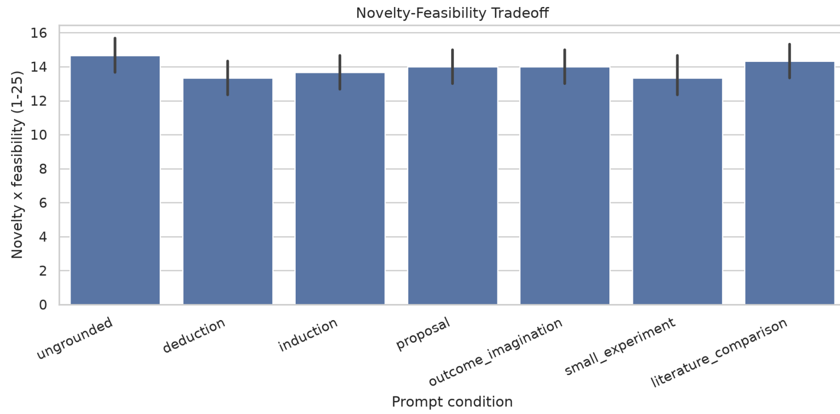

Figure 3: Novelty–feasibility product by condition. The UNGROUNDED baseline has the highest mean product, while grounded conditions do not produce a reliable gain.

this may reflect proposal fluency rather than true executable strength. LITERATURE COMPARISON often writes polished differentiation claims, yet those claims do not consistently raise judged novelty.

5 Discussion

What grounding changed. The experiment does not support the strongest version of the hypothesis. Prompt-level grounding did not significantly improve judged novelty, feasibility, specificity, overall quality, quality mean, or novelty–feasibility product. It did reliably increase lexical overlap with prior-work titles. This means the prompts made ideas more visibly connected to the supplied literature, but not reliably better under the judging rubric.

Why this matters. Literature connection is not the same as scientific novelty. In scientific ideation, a model can sound more situated by reusing terms from retrieved titles. That behavior may help readers see the local research area, but it can also narrow the model's search around the retrieved literature. The LITERATURE COMPARISON condition illustrates both sides: it has the highest mean overall score, but also the highest prior-work overlap and no reliable gain over UNGROUNDED.

Why the small experiment failed. The SMALL EXPERIMENT condition used a real audit, but the audit was shallow. It counted title terms and missing evaluation cues rather than executing code, testing claims, or checking full-paper evidence. The condition's weak novelty and overall scores suggest that "small experiment" prompts help only when the experiment produces information that changes the idea, not when it merely decorates the prompt with empirical language.

Implications for ideation systems. These results point to stronger mechanisms than prompt wording alone. A grounded ideation system should control retrieval quality, compare ideas against full abstracts or paper sections, penalize direct recombination, generate multiple diverse candidates, and calibrate automatic judges against human labels or benchmarks such as RINOBENCH. For high-value ideas, the system should also run partial execution checks or small validation experiments rather than relying on textual feasibility.

Limitations. This is a compact pilot with 12 contexts and one idea per condition per context. The judge uses coarse integer scores, producing ceiling effects around 3–4. The evaluation relies on an LLM judge, although the judge model differs from the generator. Prior-work grounding uses titles rather than full paper text, citation graphs, or expert-curated related work. The overlap metric is lexical and cannot distinguish useful technical reuse from copying. Finally, the ideas are not executed, so feasibility may be overestimated.

Broader impact. Automated ideation tools can help researchers explore directions faster, but they can also create overconfident proposals that appear literature-aware without being genuinely novel.

We therefore view grounded generation as a tool for candidate exploration, not as a replacement for expert literature review, execution, or peer evaluation.

6 Conclusion

We tested whether grounding novelty through prompt-level operations improves LLM-generated scientific ideas. In a matched pilot with 12 AI IDEA BENCH 2025 contexts and 84 generated ideas, grounding did not significantly improve judged quality metrics over an UNGROUNDED novelty prompt. The clearest effect was lexical: grounded prompts increased overlap with prior-work titles, especially under explicit literature comparison.

The main takeaway is direct: simple prompt-level grounding can make ideas look more connected to literature without making them reliably better. Future work should move from shallow prompt operations to full-text retrieval, anti-copying objectives, candidate diversification, human-calibrated judging, and execution-based validation.

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